

JOSÉ CARLOS CHAPARRO

Software Engineer · Backend & Infrastructure

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Chihuahua, MX · Open to remote / US relocation · **US Citizen (no sponsorship required)** · Spanish (native) / English (B2, upper-intermediate)

Professional Summary

Software engineer focused on backend systems, data and the infrastructure they run on. Designed and built Moneta Market's production platform, six containerized services on AWS EC2 with a 25-table PostgreSQL schema on RDS, and led the team's two interns after the lead developer left. Previously extended a VB.NET ERP used by 1,750+ employees across 170+ retail branches, including its SQL Server stored procedures. Self-hosts client products on Proxmox, Docker and Cloudflare Tunnels.

Technical Skills

Programming Languages: Python, C#, SQL, TypeScript, JavaScript

Backend: FastAPI, NestJS, Express (Node.js), Django, .NET Framework

Databases: PostgreSQL, SQL Server, SQLite

Cloud & DevOps: AWS (EC2, RDS, S3, IAM, Secrets Manager), Docker, Docker Compose, Nginx, Cloudflare, Dokploy, Backblaze B2, Git, GitHub Actions, GitLab CI/CD, Linux

Frontend: React, Next.js, Tailwind CSS, React Native, Avalonia (desktop)

Computer Vision & Edge: Ultralytics YOLO, TensorRT, ONNX, OpenCV, NVIDIA Jetson

Work Experience

Moneta Market

August 2025 – Present

Full Stack Engineer

- Replaced manual deployments with the company's first containerized platform: 6 services on AWS EC2 (Linux) run with Docker Compose (2 FastAPI microservices, Label Studio, MLflow, Nginx reverse proxy).
- Designed the 25-table PostgreSQL schema on AWS RDS that the operation runs on (24 foreign keys, 34 indexes, idempotent migrations, soft-delete audit trail), reachable only through an SSH tunnel via an EC2 bastion host.
- Built an event-driven FastAPI webhook between Label Studio and S3 with atomic DB/S3 rollback, async connection pooling and an in-memory cache with DB fallback; wrote the team runbooks.
- Led the team's two interns after the lead developer left, having co-supervised them before.
- Set up least-privilege IAM, AWS Secrets Manager, LocalStack for local S3 and the CI pipelines.
- Built the data pipeline's operator tooling (CLIs, idempotent S3-PostgreSQL sync, drift audits) and automated model training via GitLab CI on ephemeral AWS GPU instances with MLflow tracking.
- Built an on-device vision pipeline on NVIDIA Jetson Orin Nano that counts products taken from a retail cooler (TensorRT FP16, 60 fps tracking, SIFT camera recalibration); its evaluation harness set the production frame-rate and image-format requirements.
- Migrated the frame-review desktop app from Windows-only WPF to cross-platform C# (.NET 10, Avalonia): MVVM, two-pass verification, sliding-window prefetch cache.

Bifrost Code

February 2025 – Present

Co-founder & Full Stack Engineer

independent software studio; side project in my own time, alongside the full-time Moneta Market role

- Self-host every client app on the studio's own infrastructure as a low-cost alternative to Vercel/AWS: Proxmox VM (Ubuntu Server), Dokploy, Docker Compose health checks and Cloudflare Tunnels.
- Ship and run three client products in production (FisioApp, Feriu, Karaoke Balu) end to end on TypeScript pnpm monorepos: React, Next.js, NestJS, Express, PostgreSQL. Earlier projects shipped with FastAPI, Django and React Native.

Grupo Roga

March 2024 – August 2025

.NET Software Developer

- Maintained and extended a mission-critical VB.NET ERP used by 1,750+ employees across 170+ branches (Mueblería El Pasito / Villarreal Muebles / Dinexpress): POS, inventory, pricing and accounts receivable on .NET Framework.
- Designed SQL Server stored procedures for multi-branch inventory aggregation and pricing, and automated HTML report emails (Chilkat), eliminating manual report compilation for operations.
- Designed and built from scratch the ERP module for the field agents who visit customer homes for credit approval, with DevExpress maps plotting assignments and data from a companion Android app.

Selected Projects

Bifrost Code side projects, running in production

- FisioApp:** multi-tenant clinic management (patients, clinical records, appointments, payments); Next.js + PostgreSQL, WCAG 2.1 AA audited, automated off-site backups via cron to Backblaze B2.
- Feriu:** multi-tenant shared-expense platform (Splitwise-style settlement engine); NestJS + Next.js monorepo, PostgreSQL.
- Karaoke Balu:** real-time karaoke request platform (SSE-synced queue, kiosk mode, Virtual DJ integration); Express + React.

Education

Autonomous University of Chihuahua

2018 – 2023

Bachelor of Science in Computer Science Engineering

GPA 9.2/10 (Universidad Autónoma de Chihuahua); degree awarded by academic merit